Ibn Tofail University

Analysis IV

Problem Set 2

2025-2026

Exercise 1:

Study the nature of convergence of each of the following sequences of functions:

1. $f_n(x) = \frac{1}{1+n(nx-1)^2}$, for $x \in I = [0, 1]$.
2. $f_n(x) = \begin{cases} (n-1)x & \text{if } x \in \left[0, \frac{1}{n}\right], \\ 1 - x & \text{if } x \in \left[\frac{1}{n}, 1\right]. \end{cases}$
3. $f_n(x) = e^{-nx} \cos x^2$, for $x > 0$.

Correction

Exercise 2:

Consider the sequence of functions $f_n : [-1, 1] \to \mathbb{R}$, defined by

$$x \mapsto f_n(x) = e^{-nx^2} \sin nx + \sqrt{1 - x^2}, \quad n \in \mathbb{N}.$$

1. Show that the sequence of functions (f_n) converges pointwise on $I = [-1, 1]$ to a function f, which you will determine.
2. Show that (f_n) converges uniformly to f on $[\omega, 1]$, where ω is a positive constant.
3. Show that (f_n) does not converge uniformly to f on $[0, 1]$.

Correction

Exercise 3:

Determine the domain of convergence D of each series:

1. $\sum_{n>0} \frac{\cos nx}{n!}$.
2. $\sum_{n>1} \frac{1}{(z-n)^2}$.
3. $\sum_{n>1} \left|\sin \frac{x}{2^n} - \tan \frac{x}{2^n}\right|^{\frac{1}{2}}$.
4. $\sum_{n>1} \frac{n!(x-3)^{2n}}{n^{n+1}}$.

Correction

Exercise 4:

Consider the following series of functions:

$$g_n(x) = \frac{\sin x}{x^2+1} + \frac{\sin 2x}{x^2+2^2} + \ldots + \frac{\sin nx}{x^2+n^2} + \ldots$$

1. Determine the domain of convergence D.
2. Study the uniform convergence on D.

Correction

Exercise 5:

Consider the series of functions with general term:

$$f_n(x) = \frac{nx}{1+n^2x^2} - \frac{nx+x}{1+n^2x^2+2n^2x+x^2}$$

1. Determine the domain of convergence D.
2. Calculate the sum of the series.
3. What is the nature of convergence of this series on D?
4. Determine the nature of convergence of this series on $I =]0, +\infty[$ and $I = [a, +\infty[$ with $a > 0$.

Correction

Exercise 6:

Consider the series of functions with general term: $f_n(x) = (\sin x)^\alpha (\cos x)^n$.

1. Determine the sum and domain of convergence of this series.
2. Show that we have uniform convergence on any compact $[a, b]$ included in $\left]0, \frac{\pi}{2}\right[$.

Correction